CAPM and APT Models

Equilibrium and Arbitrage Approaches to Expected Return

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1 Capital Asset Pricing Model (CAPM)

The Capital Asset Pricing Model (CAPM) is a fundamental model in finance that describes the relationship between systematic risk and expected return for assets, particularly stocks.

1.1 CAPM Assumptions

1. **Homogeneous expectations.** All investors have identical expectations about expected returns, variances, and covariances of all assets.
2. **Mean-variance optimisation.** All investors are mean-variance optimisers who choose portfolios based solely on expected return and variance.
3. **Single-period model.** All investors have the same single-period investment horizon.
4. **Risk-free asset.** There exists a risk-free asset with known return r_f, and investors can borrow and lend unlimited amounts at this rate.
5. **Perfect capital markets.** No transaction costs; no taxes; perfect divisibility of assets; perfect liquidity; no restrictions on short selling.
6. **Price-taking behaviour.** All investors are price takers; no single investor can influence market prices.
7. **Unlimited borrowing and lending.** Investors can borrow and lend unlimited amounts at the risk-free rate.

1.2 CAPM Model

Under the CAPM assumptions, the expected return of any asset i is given by

$$E[R_i] = r_f + \beta_i \left(E[R_m] - r_f\right) \tag{1}$$

where

- $E[R_i]$ — expected return of asset i;
- r_f — risk-free rate of return;
- β_i — beta of asset i, measuring systematic risk;
- $E[R_m]$ — expected return of the market portfolio. Under the CAPM assumptions the market portfolio is the capitalisation-weighted portfolio of all investable assets; in practice it is typically proxied by a cap-weighted index such as the S&P 500;
- $(E[R_m] - r_f)$ — market risk premium.

The beta coefficient is defined as

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)} = \frac{\sigma_{i,m}}{\sigma_m^2}, \tag{2}$$

where $\sigma_{i,m}$ is the covariance between asset i and the market portfolio, and σ_m^2 is the variance of the market portfolio.

2 Arbitrage Pricing Theory (APT)

The Arbitrage Pricing Theory (APT), developed by Stephen Ross, provides a multi-factor approach to asset pricing that generalises the single-factor CAPM.

2.1 APT Assumptions

The APT rests on more relaxed assumptions than the CAPM.

1. **Factor structure.** Asset returns are generated by a linear factor model

$$R_i = E[R_i] + \sum_{j=1}^{k} \beta_{ij} F_j + \epsilon_i, \tag{3}$$

   where the F_j are systematic factors and ϵ_i is idiosyncratic risk.

2. **Diversifiable idiosyncratic risk.** The idiosyncratic risk ϵ_i can be diversified away in large portfolios: $E[\epsilon_i] = 0$ for all i; $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$; and $\mathrm{Cov}(\epsilon_i, F_j) = 0$ for all i, j.
3. **No arbitrage.** No arbitrage opportunities exist in the market.
4. **Perfect capital markets.** Similar to the CAPM, but less restrictive regarding investor behaviour and expectations.
5. **Finite number of factors.** There are k systematic factors driving asset returns, where k is typically much smaller than the number of assets.

2.2 APT Model

Under the APT assumptions and the no-arbitrage condition, the expected return of asset i is

$$E[R_i] = r_f + \sum_{j=1}^{k} \beta_{ij} \lambda_j \tag{4}$$

where

- $E[R_i]$ — expected return of asset i;
- r_f — risk-free rate of return;
- β_{ij} — factor loading (sensitivity) of asset i to factor j;
- λ_j — risk premium for factor j;
- k — number of systematic factors.

The factor model for returns is

$$R_i = E[R_i] + \sum_{j=1}^{k} \beta_{ij} F_j + \epsilon_i, \tag{5}$$

where F_j is the realisation of factor j (mean zero) and ϵ_i is the idiosyncratic error term for asset i.

Special case. When $k = 1$ and the single factor is the market return, the APT reduces to the CAPM.

3 Key Differences Between CAPM and APT

Aspect	CAPM	APT
Risk factors	Single (market)	Multiple factors
Assumptions	More restrictive	More flexible
Theoretical foundation	Equilibrium model	Arbitrage-based
Testability	More testable	Less testable
Practical implementation	Simpler	More complex
Market portfolio	Required	Not required

Both models provide frameworks for understanding the relationship between risk and expected return, with the APT offering greater flexibility through its multi-factor approach while the CAPM provides clearer theoretical foundations and easier implementation.